

NH3Bench: Language Agents as Real-Time Operators of an Industrial Plant

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<https://github.com/nicl-nno/NH3Bench>

Abstract

We demonstrate NH3Bench, an executable benchmark for a general question: can a language-model agent hold real-time supervisory control of an industrial installation? Three properties define the task and are absent from digital-native agent benchmarks: the process evolves whether or not the agent acts (inaction ends in catastrophe); it does not pause for deliberation (reasoning tokens convert to virtual seconds, so a correct answer can arrive after the rupture); and it is only partially digitalized (part of the truth exists only on local gauges, fetched by dispatching simulated workers). NH3Bench instantiates the task on a physics-based digital twin of an industrial ammonia (R717) refrigeration plant, with the agent as shift engineer. Across six calibrated scenarios, four Claude-family models score between 32 and 91 of 100. Attendees watch live episodes with the models’ reasoning streams and play the same scenarios themselves, under the same clock, in a browser trainer running the same twin.

Introduction

Language-model agents are being proposed for operational roles in physical industries. Beneath the specific proposals lies a general question: can an agent hold *real-time supervisory control* of a running industrial installation—keep a continuous physical process inside its safe envelope using the plant’s instruments, a fixed set of actuators and safety systems, and its human workforce, on the plant’s own clock?

Existing agent benchmarks evaluate digital-native surrogates of this task: tool use in software environments (Liu et al. 2023), customer-service dialogues under rule compliance (Yao et al. 2024), or procedure following where the correct procedure is supplied in the prompt (Nandi et al. 2025). Industrial-asset suites stop short of a closed control loop (Patel et al. 2025), and real-time operator studies use abstract text simulators without physics (Lee et al. 2026). What separates the general task from all of these is not the vocabulary of valves and compressors but three structural properties. The dynamics are autonomous: in a developing incident *doing nothing is itself a decision* with physical consequences, so there is no safe default from which to grade only errors of commission. Deliberation is not free: the plant integrates forward while the agent thinks, so correctness and

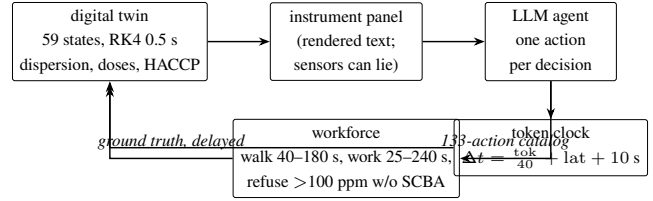


Figure 1: The NH3Bench loop. The agent sees rendered instrument readings, not plant state, and answers with one action id from a fixed 133-action catalog: remote commands, safety systems, or work orders that dispatch one of two simulated operators. Manual measurements return ground truth, bypassing failed sensors—in most scenarios SCADA disagrees with reality at least once. Every reasoning token advances the plant by 1/40 s before the action executes, so verbosity is a safety variable. A demonstration video will be available at the repository URL above.

latency cannot be scored separately. And no SCADA system sees everything—the true information set includes sight glasses, local gauges, smells and sounds, extended only by dispatching people at a cost in minutes and, sometimes, in exposure. A benchmark for the task must operationalize all three; to our knowledge none did.

NH3Bench is an executable instantiation of the task. Its three properties only bite when consequences are produced by physics rather than by script, so instantiation requires a concrete plant; we chose industrial ammonia (R717) refrigeration for its documented accident record and its rich split between digital and analog truth. The 2010 Millard Refrigerated Services ammonia release was caused by a defrost cycle interrupted by a power loss (U.S. Chemical Safety and Hazard Investigation Board 2015)—the mechanism reconstructed in our first scenario. Every outcome is produced by a thermodynamic model whose mass balance closes to under 0.1 % and which reproduces a Millard-class hydraulic shock (201 bar against a 186 bar pipe limit).

Benchmark Design

The task interface is domain-independent: the agent sees rendered instrument readings, answers with one action id from a fixed catalog, and shares its clock with the plant (Fig-

policy	S1	S2	S3	S4	S5	S6	mean
oracle (scripted)	100	100	100	100	100	100	100.0
Fable 5	100*	100*	100	100*	78	69	91.2
Opus 5	100	100	80	100	67	0	74.6
always-ESD	73	95	57	95	57	0	62.8
Sonnet 5	0	0	80	53	84	69	47.6
written regulation	0	95	80	0	94	0	44.9
Haiku 4.5	0	0	100	0	94	0	32.4
inaction	0	0	80	0	100	0	30.1

Table 1: Unified score, seed 1. * assumed, not measured; assumed cells are excluded from all physical metrics. Complete per-decision transcripts of all 21 measured model episodes are published in the repository.

ure 1). The instantiation fills this interface with physics: the twin models a 250 t/day dairy plant with four screw compressors on two suction stages, two evaporative condensers, three vessels, six air coolers, hot-gas defrost, an ice-water circuit, HACCP-critical product temperatures, and a dispersion model with per-operator dose accounting.

Six scenarios are admitted by a calibration quartet plus an always-ESD baseline: inaction must end in catastrophe; a scripted oracle must pass cleanly using only observable evidence; a faithful checklist policy (π_{reg}) must fail or pay heavily; random action must not save. Scenario mechanisms include a hung defrost whose remote abort is silently swallowed by the stuck sequencer (S1), a loud leak masking a dead level sensor (S2), an air-in-condenser false trail (S3), and an isolation trap where the regulation’s own reflex causes the rupture (S4). Two scenarios are *mirrored controls*: S5 punishes over-reaction to a drifting gas detector, while S6 presents the same evidence picture and then lets the discredited instrument start telling the truth—the fatal error is closing the question and never re-opening it. S6 also closes a loophole we found empirically: an “emergency-stop everywhere” meta-policy scores 88.6/100 on S1–S5 with no diagnosis at all, but ends in catastrophe on S6, where the leak is independent of the compressors.

The metrics attach to the task, not the domain. Alongside Prevention Rate the benchmark reports *Regulation Gap*: score minus that of the published checklist policy—an agent that cannot beat the written regulation scores below zero regardless of raw prevention. A unified 0–100 score multiplies independent axes: catastrophe or an incapacitated worker zeroes the run; operator dose multiplies (never monetized); economics and discipline fill the remainder. Points of no return, found by bisection over delayed-oracle starts, separate “did not understand” from “understood too slowly.”

Measured Results

Table 1 summarizes four Claude-family models against the reference policies. Three findings organize the demonstration. (1) *The token clock dominates the small model*: Haiku 4.5 spends $\sim 4,300$ tokens per decision, gets five decisions in S1’s 614-second window, and loses scenarios whose traps its own transcript shows it understood; in S6 it reached the correct decision—evacuate—and spent 6,598 tokens (165

virtual seconds) writing it down while the gland let go. (2) *Frontier failures are dispositional, not epistemic*: every model found the correct answer in S5 (the stronger the model, the earlier), and every model kept escalating afterwards; rankings on the mirrored pair S5/S6 invert, so no fixed disposition survives both. (3) *Choreography is a capability*: Opus 5 diagnosed S6 correctly and ordered the correct isolation three times—the first cancelled by its own evacuation order, the second refused because it had put breathing apparatus on the wrong operator, the third 232 seconds too late. The full four-model, 21-episode matrix cost $\approx \$180$ at list prices; one episode costs $\$0.3$ – 11 depending on the model.

Demonstration

After a one-minute walkthrough of the plant and the action catalog, the demonstration proceeds in three phases:

1. **Watch**: the attendee picks a model and a scenario and follows the live episode—instrument panel, virtual clock, work-order tracking, and the model’s own reasoning stream, including signature moments such as dying mid-thought with the right answer in hand.
2. **Play**: a browser trainer executes the *same* digital twin via Pyodide—no installation, physics identical by construction, a pixel-art plant view showing only what agents can see—and the attendee runs the same scenario under the same clock, with wait actions from 10 s to 60 min and click-to-inspect instrument histories. Their unified score joins a live leaderboard seeded with Table 1; the mirrored pair S5/S6 is as unforgiving of human dispositions as of model ones.
3. **Inspect**: annotated per-decision transcripts of the measured episodes, aligned with the plant trajectory, for unhurried reading.

The trainer was built for expert validation by a practicing ammonia-plant operator—the pixel map deliberately displays only agent-visible state, so humans and models play the same partially observed game.

Conclusion

NH3Bench turns a general question—can a language agent hold real-time supervisory control of an industrial installation?—into a measurable one. The capability triad it isolates belongs to the task, not to ammonia: acting under irreversible physical consequences, budgeting one’s own deliberation, and buying information with human exposure. The recipe—a calibrated physics twin, a token clock, a work-force layer, accident-forcing scenario admission with mirrored controls—transfers to any installation with a faithful twin; ammonia refrigeration is one point in that space, chosen for its accident record. At minutes and cents per episode the benchmark is cheap enough to re-run at every model release. All code, scenarios, reference policies, metric definitions and complete transcripts are public at the repository above.

References

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